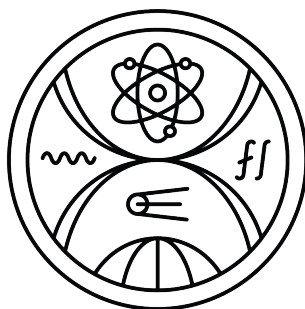


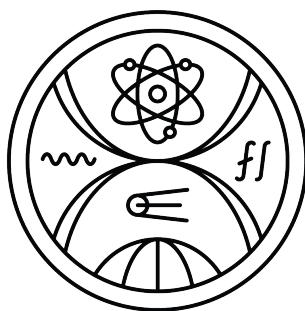
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



EXPLOITING ARTIFICIAL INTELLIGENCE IN SOFTWARE DEVELOPMENT

Master thesis

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS PHYSICS AND INFORMATICS



EXPLOITING ARTIFICIAL INTELLIGENCE IN SOFTWARE DEVELOPMENT

Master thesis

Study program: Applied informatics
Branch of study: Applied informatics
Department: Department of Applied Informatics
Supervisor: Ing. Lukáš Radoský



Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

Name and Surname: Bc. Kristián Rusnák
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Exploiting artificial intelligence in software development

Annotation: The advent of language models sparked great interest in their use in software development. Their ability to generate source code attracted particular attention. After the initial enthusiasm, attention is shifting more towards their use in other phases of the software development life cycle (SDLC), such as analysis, design, testing, and maintenance.

Choose one or more software development subtasks. Explore the possibilities of using artificial intelligence in these subtasks, especially the use of language models. Implement a solution for the selected subtask or subtasks based on artificial intelligence. Evaluate your solution on a suitable sample of data or users. Compare with existing solutions.

Aim: Design and implementation software development task solutions based on artificial intelligence.

Literature: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. <https://doi.org/10.1145/3695988>
[2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. <https://doi.org/10.1109/ICSE55347.2025.00112>
[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLOP '23). The Hillside Group, USA, Article 5, 1–31.

Keywords: artificial intelligence, large language models, software engineering, software development life cycle, SDLC

Supervisor: Ing. Lukáš Radoský
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Assigned: 14.11.2025

Approved: 17.11.2025

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ZADANIE ZÁVEREČNEJ PRÁCE

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Študijný odbor: informatika
Typ záverečnej práce: diplomová
Jazyk záverečnej práce: anglický
Sekundárny jazyk: slovenský

Názov: Exploiting artificial intelligence in software development
Využitie umelej inteligencie vo vývoji softvéru

Anotácia: Nástup jazykových modelov vyvolal veľký záujem o ich využitie vo vývoji softvéru. Veľkú pozornosť priťahovala najmä ich schopnosť generovať zdrojový kód. Po úvodnom nadšení sa pozornosť presúva viac k ich využitiu v iných fázach cyklu vývoja softvéru (SDLC), ako je analýza, návrh, testovanie, či údržba.

Zvoľte si jednu či viacero podúloh vývoja softvéru. Preskúmajte možnosti využitia umelej inteligencie pri týchto podúlohách, najmä však využitie jazykových modelov. Implementujte riešenie zvolenej podúlohy či podúloh využívajúce umelú inteligencia. Svoje riešenie vyhodnoťte na vhodne zvolenej vzorke dát alebo používateľov. Porovnajte s existujúcimi riešeniami.

Cieľ: Návrh a implementácia riešení problémov vývoja softvéru pomocou umelej inteligencie.

Literatúra: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. <https://doi.org/10.1145/3695988>
[2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. <https://doi.org/10.1109/ICSE55347.2025.00112>
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Kľúčové slová: umelá inteligencia, jazykové modely, softvérové inžinierstvo, cyklus vývoja softvéru

Vedúci: Ing. Lukáš Radoský
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Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.



Univerzita Komenského v Bratislave
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Spôsob sprístupnenia elektronickej verzie práce:
bez obmedzenia

Dátum zadania: 14.11.2025

Dátum schválenia: 17.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce

I hereby declare that I have authored this thesis independently, using only the referenced sources and literature, and under the guidance of my thesis advisor. All external contributions have been properly acknowledged.

Bratislava, 2026

.....
Bc. Kristián Rusnák

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I would like to express my sincere gratitude to Ing. Lukáš Radoský for his continuous support, insightful advice, and expert guidance throughout the development of this thesis. His deep knowledge in the field of artificial intelligence and software development was instrumental in shaping this work. I would also like to thank my partner and my family, whose encouragement and patience throughout my academic journey meant more than words can express.

Abstract

Content of abstract.

Keywords: keywords 1, keywords 2, keywords 3

Abstrakt

Obsah abstraktu.

Klíčové slová: Klíčové slov 1, Klíčové slov 2, Klíčové slov 3

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Terminology

Terms

- **Term 1**
Explanation 1.
- **Term 2**
Explanation 1.
- **Term 3**
Explanation 1.

Abbreviations

- **AAA** - Abbreviations 1.
- **BBB** - Abbreviations 2.
- **CCC** - Abbreviations 3.

Motivation

Motivation

Chapter 1

Introduction

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Introduction 1

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Conclusion

Conclusion

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- [3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. A prompt pattern catalog to enhance prompt engineering with ChatGPT. *arXiv preprint arXiv:2302.11382*, 2023. Submitted Feb 2023.
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